

Question number	Answer	Mark
4(a)(i)	D hydrolysis	(1)

Question number	Answer	Mark
4(a)(ii)	A calculation in which: $\text{kJ of energy} = \text{mass of lactose in 200 g of milk} \times \text{energy per gram}$ Example of calculation: $4.9 \times 2 \times 16 = 156.8 \text{ kJ}$	(1)

Question number	Answer	Additional guidance	Mark
4(b)	An answer that includes the following points: similarities: • both polysaccharides / formed from many (alpha) glucose monomers (1) • joined by glycosidic bonds (1) differences: • starch is composed of two polysaccharides, glycogen only one (1) • amylose is a {straight / helical} chain, amylopectin and glycogen are both branched molecules (1)	Allow description of glycosidic bonds as 1,4 and 1,6	(4)

Question number	Answer	Additional guidance	Mark
4(c)(i)	An answer that includes one of the following points: • saturated lipids have no carbon – carbon double bonds and unsaturated lipids have a carbon – carbon double bond or • saturated lipids have straight chains and unsaturated lipids have bent chains or • saturated lipids have a greater ratio of hydrogen to carbon / unsaturated lipids have a lower ratio of hydrogen to carbon	Answer must be comparative	(1)

Question number	Answer	Mark
4(c)(ii)	An description that includes the following points: • one glycerol molecule and three fatty acid molecules (1) • which react via a condensation reaction (1) • and are joined by ester bonds (1)	(3)